Results appendix — SYNTHETIC POSITIVE CONTROL (planted edge)

BTC/USDT 4h bars, 2019-09-01 to 2026-10-26. Walk-forward from 2022-01-01 with 21 quarterly refits; purge + 6-bar embargo. Test year (untouched during design): from 2025-10-27. Costs: 0.26% per round trip. Chosen model by the pre-registered rule: score. Edge validated on dev (kill switch): YES.

1. Does the model predict the trade outcome? (out-of-sample)

Model	Period	AUC target	AUC stop	Log-loss	Brier	IC exp. R
Base rate	dev	0.490	0.492	1.0125	0.6126	-0.017
Base rate	test	0.465	0.474	0.9977	0.6039	-0.109
Logistic	dev	0.620	0.570	0.9975	0.6054	0.145
Logistic	test	0.629	0.565	0.9818	0.5948	0.110
GBM (depth 3)	dev	0.595	0.546	1.0017	0.6065	0.102
GBM (depth 3)	test	0.619	0.561	0.9846	0.5963	0.079

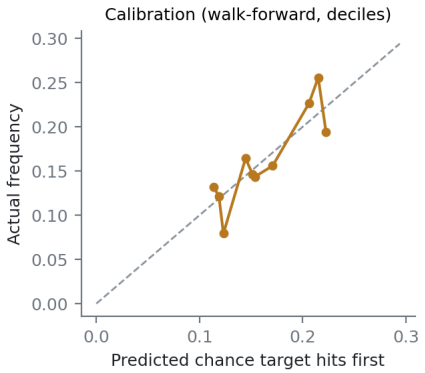
Outcome mix in the test year: target first 15.0%, stop first 34.5%, timed out 50.5%. IC = Spearman rank correlation between the model's expected result and the realised net result per bar.

2. Trading results, net of costs (one position at a time, 100% of account, no leverage)

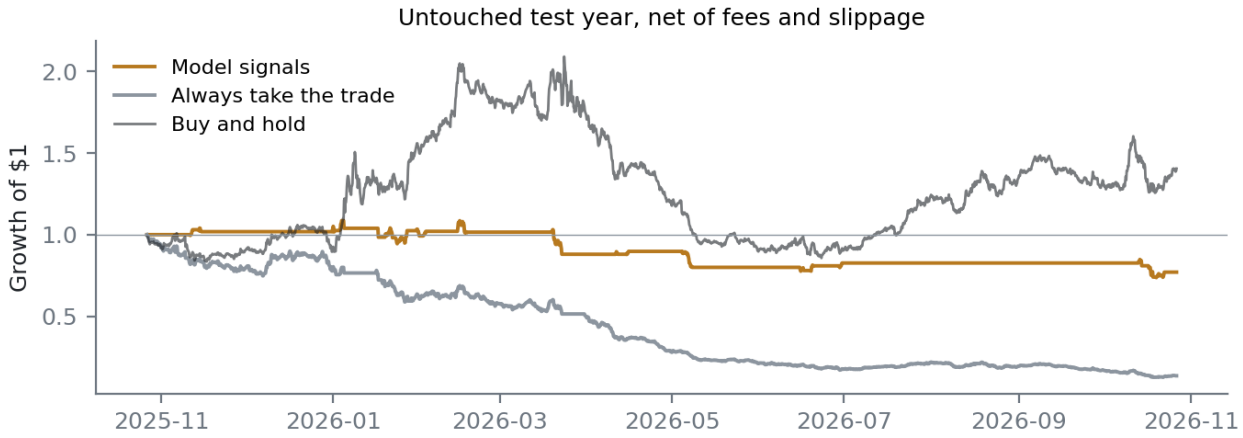
Strategy	Period	Trades	Win	Stop	Avg net R	CAGR	Sharpe	Max DD	Exposure
Model: all signals	dev	121	20.7%	20.7%	0.20	23.4%	1.30	-13.8%	7.1%
Model: all signals	test	42	7.1%	35.7%	-0.17	-23.1%	-1.05	-31.8%	8.9%
Model: Medium+ only	dev	0	–	–	–	0.0%	0.00	0.0%	0.0%
Model: Medium+ only	test	0	–	–	–	0.0%	0.00	0.0%	0.0%
Always take trade	dev	1695	17.5%	35.9%	-0.12	-74.7%	-2.26	-99.5%	93.2%
Always take trade	test	432	15.5%	36.1%	-0.16	-86.7%	-3.94	-87.1%	92.1%
Buy and hold	dev	–	–	–	–	-8.1%	0.17	-74.1%	100.0%
Buy and hold	test	–	–	–	–	40.7%	0.86	-58.7%	100.0%

Configurations tried: 8. Deflated Sharpe probability (dev): 0.792 (values near 1 mean the Sharpe is unlikely to be luck from trying several variants).

Costs	Model trades	Model CAGR	Model Sharpe	Always CAGR	Always Sharpe
0.0x	42	-14.1%	-0.58	-58.7%	-1.61
1.0x	42	-23.1%	-1.05	-86.7%	-3.94
2.0x	42	-31.1%	-1.50	-95.7%	-6.21









3. Honest track record by confidence tier (test year)

Tier	Signal bars	Target first	Stop first	Predicted R	Realised net R
Low	127	9.4%	32.3%	0.06	-0.11
Medium	0	–	–	–	–
High	0	–	–	–	–

Share of test-year bars with no signal: 93.8%.

4. Robustness

Check	Period	AUC target	IC	Trades	Sharpe	Always Sharpe
Barriers ×0.75	dev	0.586	0.141	132	0.94	-3.19
Barriers ×0.75	test	0.570	0.095	29	-0.51	-4.79
Barriers ×1.25	dev	0.573	0.151	22	0.76	-1.28
Barriers ×1.25	test	0.525	0.059	0	0.00	-1.99
Drop 'taker'	dev	0.528	-0.002			
Drop 'taker'	test	0.581	-0.000			

Year	AUC target	IC exp. R	Signal coverage	Signal avg net R
2022	0.576	0.135	0.3%	0.49
2023	0.572	0.121	5.2%	0.35
2024	0.595	0.165	2.8%	0.15
2025	0.549	0.112	10.1%	0.05
2026	0.620	0.069	6.9%	-0.13
Regime: calm	0.570	0.127		
Regime: normal	0.579	0.133		
Regime: stormy	0.647	0.066		

Volatility model (QLIKE, lower is better): HAR 0.405 vs EWMA 0.432 on dev; HAR 0.406 vs EWMA 0.424 on test.

Largest logistic weights (target-vs-stop, standardised): taker +0.510, ema_slope -0.204, dist_hi_7d +0.148, park_24 -0.130, ret_24 -0.126, vol_ratio +0.120.

6. Data integrity

Source: synthetic. Raw rows None, clean rows None, duplicate-timestamp rows None (corrupted duplicates kept only if consistent with neighbours), missing 4h bars None (exchange outages, all before March 2020), labels

dropped because their 24h window crossed a gap: 0. Every 4h candle was cross-checked against the independent 1h file: all but one corrupted 1h row agree within 0.2%.